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# GD32F303RCT6 Firmware Build And Flash Record ## Goal Build a burnable firmware image
on macOS without Keil and without modifying `GD32F303RXT6 KIT`. ## Target - Board SDK:
`GD32F303RXT6 KIT/3. SDK/GD32F303RCT6 Demo` - MCU class: GD32F30x high-density,
Cortex-M4F - Memory map: 256 KiB Flash at `0x08000000`, 48 KiB usable SRAM at
`0x20000000` - ST-Link probe result: `V2J37S7`, flash `262144`, SRAM `65536`, chipid
`0x414`, dev-type `F1xx_HD` - SRAM note: ST-Link reports 64 KiB through the STM32F1-
compatible path, but the vendor Keil projects declare `IRAM(0x20000000,0x0C000)`, so the
linker script uses 48 KiB. ## Toolchain Installed with Homebrew: ``sh brew install cmake ninja
arm-none-eabi-gcc stlink open-ocd `` Verified versions: ``sh arm-none-eabi-gcc --version cmake
--version ninja --version st-info --version openocd --version `` ## Architecture - `firmware/
CMakeLists.txt`: GCC/CMake build definition - `firmware/cmake/arm-none-eabi-gcc.cmake`:
cross toolchain file - `firmware/ld/GD32F303RCT6.ld`: 256 KiB Flash / 48 KiB SRAM linker
script - `firmware/startup/startup_gd32f30x_hd_gcc.S`: GNU startup and vector table -
`firmware/src/main.c`: validation firmware loop - `firmware/src/board.c`: RGB LED and
USART0 board support - `firmware/src/systick.c`: millisecond timebase - `firmware/include/
stdint.h` and `firmware/include/stdbool.h`: freestanding C headers for the Homebrew Arm GCC
package, which does not ship newlib headers - `GD32F303RXT6 KIT`: read-only SDK
dependency ## Build From `...`: ``sh cmake -S firmware -B firmware/build -G Ninja
-DCMAKE_TOOLCHAIN_FILE=.../firmware/cmake/arm-none-eabi-gcc.cmake
-DCMAKE_BUILD_TYPE=Release cmake --build firmware/build `` Expected outputs: -
`firmware/build/gd32f303rct6_firmware.elf` - `firmware/build/gd32f303rct6_firmware.bin` -
`firmware/build/gd32f303rct6_firmware.hex` - `firmware/build/gd32f303rct6_firmware.map`
Current LCD diagnostic build after SRAM correction: ``text FLASH: 25484 B / 256 KiB RAM:
4240 B / 48 KiB `` ## Flash Probe: ``sh st-info --probe `` Write and reset: ``sh st-flash --reset
write firmware/build/gd32f303rct6_firmware.bin 0x08000000 `` Flash result on 2026-05-14:
``text Flash written and verified! jolly good! NRST is not connected --> using software reset via
AIRCRR `` Readback verification: ``sh st-flash read firmware/build/
gd32f303rct6_firmware_readback.bin 0x08000000 3004 cmp firmware/build/
gd32f303rct6_firmware.bin firmware/build/gd32f303rct6_firmware_readback.bin shasum -a 256
firmware/build/gd32f303rct6_firmware.bin firmware/build/gd32f303rct6_firmware_readback.bin
`` Readback SHA256: ``text
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware.bin
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware_readback.bin `` ## Runtime Validation - RGB LED pins are `PC0`,
`PC1`, `PC2`, based on the vendor RGB LED example. - The firmware toggles red, green, and
blue channels in sequence every 250 ms. - USART0 uses `PA9` TX and `PA10` RX at `115200
8N1` and emits `GD32F303RCT6 GCC firmware ready` followed by `tick` lines. -
`gpio_pin_remap_config(GPIO_SWJ_SWDPENABLE_REMAP, ENABLE)` keeps SWD
enabled after boot.

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